

Name : Preksha Ramkrushna Wagh

Roll No : 198

Title : Write a Java Program(s) that demonstrates the use of Collection Classes (Collection frame work)

```
package javaapplication29;

import java.util.ArrayList;
import java.util.Collections;
import java.util.HashMap;
import java.util.HashSet;

public class JavaApplication29 {

    public static void main(String[] args) {
        //==Array List==
        System.out.println("==ArrayList Example== : ");
        ArrayList<String> fruits = new ArrayList<>();

        fruits.add("Mango");
        fruits.add("Apple");
        fruits.add("Banana");
        fruits.add("Orange");

        System.out.println("Original List : " + fruits);

        fruits.set(1, "Pineapple");
        System.out.println("After Update : " + fruits);

        fruits.remove("Banana");
        System.out.println("After Remove : " + fruits);

        Collections.sort(fruits);
        System.out.println("After Sorting : " + fruits);

        //==HashSet
        System.out.println("\n==HashSet Example== : ");

        HashSet<Integer> numbers = new HashSet<>();

        numbers.add(40);
        numbers.add(10);
        numbers.add(30);
        numbers.add(20);

        System.out.println("Original Set : " + numbers);
        numbers.remove(30);
        System.out.println("\nAfter Remove : " + numbers);
    }
}
```

```
//==HashMap
System.out.println("\n==HashMap Example== : ");

HashMap<Integer, String> students = new HashMap<>();

students.put(1, "Pooja");
students.put(2, "Sakshi");
students.put(3, "Prajakta");

System.out.println("Original Map : " + students);

students.put(2, "Ankit");
System.out.println("\nAfter Update : " + students);

students.remove(3);
System.out.println("After Remove : " + students);

}

}
```

Output :

```
==ArrayList Example== :
Original List : [Mango, Apple, Banana, Orange]
After Update : [Mango, Pineapple, Banana, Orange]
After Remove : [Mango, Pineapple, Orange]
After Sorting : [Mango, Orange, Pineapple]
```

```
==HashSet Example== :
Original Set : [20, 40, 10, 30]
```

```
After Remove : [20, 40, 10]
```

```
==HashMap Example== :
Original Map : {1=pooja, 2=Sakshi, 3=Prajakta}
```

```
After Update : {1=pooja, 2=Ankit, 3=Prajakta}
After Remove : {1=pooja, 2=Ankit}
```

Title:-write a java program to demonstrate EJB

1.StateFull

```
package shoping;
```

```
import java.util.*;
```

```
public class bean {  
    public List <String> data= new ArrayList<>();  
    public List <Integer> data1= new ArrayList<>();  
  
    public void additems (String items)  
    {  
        data.add(items);  
    }  
    public List getitems()  
    {  
        return this.data;  
    }  
    public void additems (int items)  
    {  
        data1.add(items);  
    }  
    public List getitems1()  
    {  
        return this.data1;  
    }  
}  
package shoping;
```

```
public class Shoping {  
  
    public static void main(String[] args) {  
        bean b1=new bean();  
        b1.additems ("tree");  
        b1.additems ("flower");  
        b1.additems ("leaf");  
        b1.additems ("root");  
  
        b1.additems (30);  
        b1.additems (300);  
        b1.additems (400);  
        b1.additems (50);  
  
        System.out.println(b1.getitems());  
        System.out.println(b1.getitems1());  
    }  
}
```

```
}
```

Output

[tree ,flower ,leaf, root]

[30, 300, 400, 50]

2.StateLess

```
package cal;
public class Cal {
    public static void main(String[] args) {
        num n1=new num();
        n1.add(30,20);
        n1.sub(50,20);
        n1.mul(15,5);
        n1.div(50,5);

        System.out.println(n1.add(10,20));
        System.out.println(n1.sub(50,20));
        System.out.println(n1.mul(10,5));
        System.out.println(n1.div(50,5));
    }
}
```

```
package cal;
public class num {
    public int add(int a,int b)
    {
        return a+b;
    }
    public int sub(int a,int b)
    {
        return a-b;
    }
    public int mul(int a,int b)
    {
        return a*b;
    }
    public int div(int a,int b)
    {
        return a/b;
    }
}
```

Output:

30
30
50
10

3.Entity Bean

```
package student;
public class Student {
    public static void main(String[] args) {
        bean1 b1=new bean1();
        b1.setid(163);
        b1.setname("sanu");
        b1.setage(22);

        System.out.println(b1.getid());
        System.out.println(b1.getname());
        System.out.println(b1.getage());
    }
}
```

```
package student;
public class Student {
    public static void main(String[] args) {
        beans b1=new beans();
        b1.setid(123);
        b1.setname("Janhavi");
        b1.setage(23);

        System.out.println(b1.getid());
        System.out.println(b1.getname());
        System.out.println(b1.getage());
    }
}
```

```
package student;

public class bean1 {
    private int id;
    private String name;
    private int age;
    public void setid(int data)
    {
        this.id=data;
    }
}
```

```

public int getid()
{
    return this.id;
}
public void setname(String data)
{
    this.name=data;
}
public String getname()
{
    return this.name;
}
public void setage(int data)
{
    this.age=data;
}
public int getage()
{
    return this.age;
}
}

```

Output

```

123
Janhavi
23

```

Practical no:write java program that demonstrate use of JSP technology

INDEX

```

<html>
<head>
    <title>Admission form</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>
    <h1>Student information</h1>
    <form action="home.jsp" method="get">
        <label>Enter your name:</label>
        <input type="text" name="name" placeholder="Enter the name">
        <br><br> <label>Enter your age:</label>
        <input type="text" name="age" placeholder="Enter the age">
        <br><br>
    </form>

```

```
        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

HOME

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<body>
    <%
        String name = request.getParameter("name");
        String age = request.getParameter("age");
        if (name != null) session.setAttribute("userName", name);
        if (age != null) session.setAttribute("userAge", age);
    %>
    <div>
        <a href="index.html">Home (Form)</a> |
        <a href="about.jsp">About</a> |
        <a href="contact.jsp">Contact</a>
    </div>
    <hr>
    <marquee>
        Welcome to homepage: <%= session.getAttribute("userName") %>
        (Age: <%= session.getAttribute("userAge") %>)
    </marquee>
    <h1>Home Page</h1>
    <p> Your Data Will Be Submitted</p>

    </body>
</html>
```

ABOUT

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
    </head>
    <body>
        <div>
            <a href="index.html">Home (Form)</a> |
            <a href="home.jsp">Home Page</a> |
            <a href="contact.jsp">Contact</a>
        </div>
    <hr>
```

```

<h1>About Page</h1>
<p>Welcome To About Page:</p>
<h3>Name: <%= session.getAttribute("userName") %></h3>
<h3>Age: <%= session.getAttribute("userAge") %></h3>

</body>
</html>

```

CONTACT

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">

  </head>
  <body>

    <title>contact us</title>
  <div>
    <a href="index.html">Home (Form)</a> |
    <a href="home.jsp">Home Page</a> |
    <a href="about.jsp">About</a>
  </div>
  <hr>
  <h1>Contact Page</h1>
  <p>User Name: <%= session.getAttribute("userName") %></p>
  <p>User Age: <%= session.getAttribute("userAge") %></p>

  </body>
</html>

```

Practical no:write java program that demonstrate use of RMI technology

```

package rmi1;
import java.rmi.Naming;
import java.rmi.registry.LocateRegistry;
public class Rmi1 {
  public static void main(String[] args) {
    try{
      LocateRegistry.createRegistry(5000);
      add_imp skeleton=new add_imp();
      Naming.rebind("rmi://localhost:5000/ADD", skeleton);
      System.out.println("server ready");
    }
  }
}

```

```

        catch(Exception e){
            System.out.println("Error:" + e);
        }
    }
}

package rmi1;
import java.rmi.Remote;
import java.rmi.RemoteException;
public interface add_I extends Remote {
    public int add(int x, int y) throws RemoteException;
}

package rmi1;
import java.rmi.Naming;
import java.rmi.RemoteException;
import java.util.logging.Level;
import java.util.logging.Logger;
public class client {
    public static void main(String args[]){
        try{
            add_I stub=(add_I) Naming.lookup("rmi://localhost:5000/ADD");
            System.out.println("Result is-" + stub.add(10, 50));
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}

package rmi1;
import java.rmi.server.*;
import java.rmi.RemoteException;
public class add_imp extends UnicastRemoteObject implements add_I {
    public add_imp() throws RemoteException{
        super();
    }
    public int add(int a, int b){
        return a+b;
    }
}

```

Output:

run:

server ready

Result is-60

Title : Write a program to demonstrate java bean

1.1 package studentclient;

```
//import javax.persistence.Entity;  
//import javax.persistence.Id;
```

```
public class StudentEntityBean {  
    // @Id  
    private int id;  
    private String name;  
    private String course;  
  
    // Setter methods  
    public void setId(int id) {  
        this.id = id;  
    }  
  
    public void setName(String name) {  
        this.name = name;  
    }  
  
    public void setCourse(String course) {  
        this.course = course;  
    }  
  
    // Getter methods  
    public int getId() {  
        return id;  
    }  
  
    public String getName() {  
        return name;  
    }  
  
    public String getCourse() {  
        return course;  
    }  
}
```

1.2 package studentclient;

```
public class StudentClient {  
  
    public static void main(String[] args) {
```

```
StudentEntityBean student = new StudentEntityBean();

student.setId(101);
student.setName("Pooja");
student.setCourse("MCA");

System.out.println("Student ID: " + student.getId());
System.out.println("Student Name: " + student.getName());
System.out.println("Course: " + student.getCourse());
}

}
```

Output :

Student ID: 101
Student Name: Pooja
Course: MCA

#ArrayList

```
package collection;

import java.util.Scanner;

import java.util.ArrayList;

import java.util.List;

public class Collection {

    public static void main(String[] args) {

        List obj = new ArrayList<>();

        obj.add("Kunika ");

        System.out.println(obj);

        obj.add(10);

        obj.add(20);

        obj.add(30);

        System.out.println(obj);

    }

}
```

```
obj.remove(0);

System.out.println(obj);

Scanner sc = new Scanner(System.in);

System.out.println("Enter the index");

int i = sc.nextInt();

obj.remove(i);

System.out.println(obj);

}
```

Practical Name: 2.1:- Write java program(s) that demonstrates generic programming.

1.

```
package demo11;

public class Generic_class <T> {

}

public void method(T[] arr) {

}

for (T element: arr) {

}

System.out.println(element);
}
}

package demo11;

public class Demo11 {

public static void main(String[] args) {

Generic_class obj = new Generic_class();
```

```
String a[] = {"ruhi","moksh"};
```

```
obj.method(a);
```

```
}
```

```
Integer b[] = {1, 2};
```

```
obj.method(b);
```

```
}
```

```
}
```

2. Single_type generic class:

```
package gen_class;
```

```
public class single_type <T> {
```

```
private T val;
```

```
this.val = value;
```

```
}
```

```
public void set_value(T value) { }
```

```
public T get_value() {
```

```
return val;
```

```
}
```

```
package gen_class;
```

```
public class Gen_class {
```

```
public static void main(String[] args) { single_type obj = new single_type();
```

```
String val = "Text Value";
```

```
obj.set_value(val);
```

```
System.out.println("String value -> " + obj.get_value());

Integer val1 = 22222;

obj.set_value(val1);

System.out.println("Integer value -> " + obj.get_value()); }

}
```

Multiple_type generic class:

```
package gen_class2;

public class multiple_type<T, U> {

    private T val1;
    private U val2;

    val1 = v1;

}

public void set_value(T v1, U v2) {
    Val1=v1;
    val2 = v2;
}

public T get_val1() {
    return val1;
}

public U get_val2() {
    return val2;
}
}

package gen_class2;

public class Gen_class2 {

    public static void main(String[] args) {
```

```

multiple_type<String, Integer> obj = new multiple_type<>();

obj.set_value("Kunika", 101);

System.out.println("Name: " + obj.get_val1());

System.out.println("Roll No: " + obj.get_val2());

}
}

```

Title: Write a java Program that Demonstrate the use of Collection Classes

1.1)

```

package collection1;
import java.util.Stack;
public class Collection1 {
    public static void main(String[] args) {
        Stack obj=new Stack<>();
        obj.push(50);
        obj.push("data");
        obj.push(250);
        obj.push(5.5);
        obj.pop();
        System.out.println(obj);
        System.out.println(obj.peek());
        System.out.println(obj.size());
    }
}

```

Output:

```

[50, data, 250]
250
3

```

1.2)a

```

package que;
import java.util.LinkedList;
import java.util.Queue;
public class que{
    public static void main(String[] args) {
        System.out.println("LinkedList Queue");
    }
}

```

```

Queue<Integer>q1=new LinkedList<>();
q1.add(10);
q1.add(20);
q1.add(30);
System.out.println("Queue Element "+q1);
System.out.println("Peek Element "+q1.peek());
if(q1.contains(20))
System.out.println("20 Found in Queue");
System.out.println("Removed Element "+q1.remove());
System.out.println("Queue After Removal "+q1);
System.out.println("Queue Size"+q1.size());
}
}

```

Output:

```

LinkedList Queue
Queue Element [10, 20, 30]
Peek Element 10
20 Found in Queue
Removed Element 10
Queue After Removal [20, 30]
Queue Size2

```

1.2)b

package prioque;

import java.util.*;

```

public class Prioque {
    public static void main(String[] args) {
        System.out.println("\nPriority Queue");
        Queue<Integer>q2=new PriorityQueue<>();
        q2.add(50);
        q2.add(10);
        q2.add(30);
        System.out.println("Queue Elements"+q2);
        System.out.println("Peek Element"+q2.peek());
        if(q2.contains(30))
            System.out.println("30 Found in Queue");
        System.out.println("Removed Element"+q2.remove());
        System.out.println("Queue After Removal"+q2);
        System.out.println("Queue Size"+q2.size());
    }
}

```

Output:

```

Priority Queue
Queue Elements[10, 50, 30]
Peek Element10
30 Found in Queue
Removed Element10

```

Queue After Removal[30, 50]
Queue Size2

1.2)c

```
package arrdeq;
import java.util.ArrayDeque;
import java.util.Queue;
public class Arrdeq {
    public static void main(String[] args) {
        System.out.println("\nArrayDeque Queue");
        Queue<Integer>q3=new ArrayDeque<>();
        q3.add(5);
        q3.add(15);
        q3.add(25);
        System.out.println("Queue Elements"+q3);
        System.out.println("Peek Elemen"+q3.peek());
        if(q3.contains(15))
            System.out.println("15 Found in Queue");
        System.out.println("Removed Element"+q3.remove());
        System.out.println("Queue After Removal"+q3);
        System.out.println("Queue Size"+q3.size());
    }
}
```

Output:

ArrayDeque Queue
Queue Elements[5, 15, 25]
Peek Elemen5
15 Found in Queue
Removed Element5
Queue After Removal[15, 25]
Queue Size2

1.2)d

```
package arrblocque;
import java.util.concurrent.ArrayBlockingQueue;
public class Arrblocque {
    public static void main(String[] args) {
        System.out.println("\nArrayBlockingQueue");
        ArrayBlockingQueue<Integer>q4=new ArrayBlockingQueue<>(5);
        q4.add(100);
        q4.add(200);
        q4.add(300);
        System.out.println("Queue Elements"+q4);
        System.out.println("Peek Element"+q4.peek());
    }
}
```

```

        if(q4.contains(200))
        System.out.println("200 Found in Queue");
        System.out.println("Removed Element"+q4.remove());
        System.out.println("Queue After Removal"+q4);
        System.out.println("Queue Size"+q4.size());
    }
}

```

Output:

```

ArrayBlockingQueue
Queue Elements[100, 200, 300]
Peek Element100
200 Found in Queue
Removed Element100
Queue After Removal[200, 300]
Queue Size2

```

1.3)a

package hs;

import java.util.HashSet;

import java.util.Set;

public class HS {

```

    public static void main(String[] args) {
        System.out.println("HashSet");
        Set<Integer>s1=new HashSet<>();
        s1.add(10);
        s1.add(20);
        s1.add(30);
        s1.add(40);
        System.out.println("Set Elements"+s1);
        if(s1.contains(20))
        System.out.println("20 Found in Set");
        System.out.println("Set After Removal"+s1);
        System.out.println("Set Size"+s1.size());
    }
}

```

Output:

```

HashSet
Set Elements[20, 40, 10, 30]
20 Found in Set
Set After Removal[20, 40, 10, 30]
Set Size4

```

1.3)b

package lhs;

import java.util.LinkedHashSet;

import java.util.Set;

public class LHS {

public static void main(String[] args) {

System.out.println("\nLinkedHashSet");

Set<Integer>s2=new LinkedHashSet<>();

s2.add(5);

s2.add(15);

s2.add(20);

s2.add(25);

System.out.println("Set Elements"+s2);

if(s2.contains(25))

System.out.println("25 Found in Set");

s2.remove(5);

System.out.println("Set After Removal"+s2);

System.out.println("Set Size"+s2.size());

}

}

Output:

LinkedHashSet

Set Elements[5, 15, 20, 25]

25 Found in Set

Set After Removal[15, 20, 25]

Set Size3

1.3)c

package ts;

import java.util.Set;

import java.util.TreeSet;

public class TS {

public static void main(String[] args) {

System.out.println("\nTreeSet");

Set<Integer>s3=new TreeSet<>();

s3.add(50);

s3.add(10);

s3.add(30);

s3.add(10);

System.out.println("Set Elements"+s3);

if(s3.contains(30))

System.out.println("30 Found in Set");

s3.remove(10);

System.out.println("Set After Removal"+s3);

System.out.println("Set Size"+s3.size());

}

```
}
```

Output:

TreeSet

Set Elements[10, 30, 50]

30 Found in Set

Set After Removal[30, 50]

Set Size2